INCH-POUND
MIL-I-24718/3
6 April 1990

MILITARY SPECIFICATION SHEET

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING POLYESTER DIALLYL PHTHALATE, SLIGHTLY THIXOTROPIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24718.

Resin: The product shall consist of primarily polyester resin with diallyl phthalate monomer, catalyst and appropriate filler and thixotropes to satisfy the requirements specified.

Thermal class: The cured resin shall be of a thermal class as required by the contracting activity (see 4.6.15.1).

Rheology: The rheological properties shall be as specified and suitable for VPI of form and random wound electrical coils and windings.

First article inspection. First article inspection shall be as specified in MIL-I-24718 and table I herein.

TABLE I. First article inspection.

Characteristics	Unit	Requirements
Catalyzed storage life	Months at 20°C, minimum	9
Vapor pressure	mm Hg, maximum	2.0
Weight loss on cure	Average percent, maximum	30.0
Flash point	°C, minimum	93
Viscosity		
at 2 r/min	Average centipoise, maximum	15,000
at 20 r/min	Average centipoise, maximum	5,000
Shear thinning index	Minimum	1.10
Build	Mils, minimum	2.0
	Mils, maximum	5.0

AMSC N/A FSC 5970

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TABLE I First article inspection - Continued.

Characteristics	Unit	Requirements
Resin retention	Weight percent, minimum	75.0
Dielectric strength		
at 96/23/50	Volts per mil, minimum average per panel	1,000
at 96/23/96	Percent decrease from 96/23/50 value, maximum	25
at 24/23/water	Percent decrease from 96/23/50 value, maximum	25
Dissipation factor		
at 23°C	Percent, maximum	3.0
Dielectric constant		
at 23°C	Maximum	5.0
Bond strength		
at 23°C	Pounds force, minimum	30.0
at 150°C	Pounds force, minimum	5.0
Thermal endurance		
Class 105	°C at 20,000 hours	105
Class 130	°C at 20,000 hours	130
Class 155	°C at 20,000 hours	155
Class 180	°C at 20,000 hours	180
Class 200	°C at 20,000 hours	200
Saltwater proof	Hours minimum	100
Hydrolytic stability	Percent decrease in average bond strength, maximum	50
Chemical resistance		
Hydraulic fluid	Percent decrease in average bond strength, maximum	25
Lubricating oil	Percent decrease in average bond strength, maximum	25
Cleaning fluid	Percent decrease in average bond strength, maximum	25
Distilled water	Percent decrease in average bond strength, maximum	25
Detergent solution	Percent decrease in average bond strength, maximum	25
Hardness	Type D, minimum	70

Quality conformance inspection. Quality conformance inspection shall be as specified in MIL-I-24718 and table II herein:

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PART II. Quality conformance requirements.

Characteristics	Unit	Requirements
Weight loss on cure	Average percent, maximum	30
Gel point	Percentage of qualification value, plus or minus	10
Viscosity at 2 r/min	Average centipoise, maximum	15,000
	Percentage of qualification value, plus or minus	10
Build	Mils, minimum	2.0
	Mils, maximum	5.0
Resin retention	Weight percent, minimum	75.0
Specific gravity	Percentage of qualification value, plus or minus	10.0
Dielectric strength at 24/23/50	Volts per mil, minimum	1,000
Hardness	Percentage of qualification value, plus or minus	10.0
Bond strength at 25°C	Pounds force, minimum	30.0

Intended use. The product shall be used for VPI of form or random wound electrical coils. It shall be used only as new resin for starting up a VPI process or replenishing stored VPI resin of the same formulation and from the same manufacturer.

Part or Identifying Number (PIN): The PIN shall be M24718/3-(dash number indicating thermal class). For example, M24718/3-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
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	MIL-I-24718/3	

DOCUMENT TITLE

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING, GENERAL SPECIFICATION FOR

NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

REASON FOR RECOMMENDATION

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a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. C.Y. Lu (SEA 56Z24) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 703-602-3123	8-332-3123
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy Washington, DC 20362-5101	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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